

Cyber Security Fundamentals

Risk, Threats, Vulnerabilities & Ports
With Analysis, Assessment and
Examples

Presentation Outline

- - Introduction to Cyber Security
- - Risk, Threats, and Vulnerabilities
- - Relationship Between Risk, Threat & Vulnerability
- - Risk Analysis and Risk Assessment
- - Network Ports and Their Working
- - Common Ports and HTTP Status Codes
- - Role of Ports in Cyber Security
- - Practical Examples

Introduction to Cyber Security

- Cyber Security is the practice of protecting systems, networks, and data from digital attacks.
- It focuses on confidentiality, integrity, and availability (CIA Triad).

Threat

- A threat is any potential danger that can exploit a vulnerability.
- Examples:
 - - Malware
 - - Phishing emails
 - - Hackers
 - - Natural disasters

Vulnerability

- A vulnerability is a weakness in a system, network, or software.
- Examples:
 - - Weak passwords
 - - Unpatched software
 - - Open ports
 - - Misconfigured firewalls

Risk

- Risk is the possibility of loss or damage when a threat exploits a vulnerability.
- $\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Impact}$

Relationship

- Threat: The attacker or danger
 - Vulnerability: The weakness
 - Risk: The potential damage
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- Example:
 - Threat: Hacker
 - Vulnerability: Weak password
 - Risk: Account compromise

Types of Threats

- - Malware (Virus, Worm, Trojan)
- - Phishing
- - DoS / DDoS Attacks
- - Insider Threats
- - Man-in-the-Middle (MITM)

Types of Vulnerabilities

- - Software vulnerabilities
- - Network vulnerabilities
- - Human vulnerabilities
- - Physical vulnerabilities

Risk Analysis

- Risk Analysis is the process of identifying risks and understanding their impact.
- Steps:
 - 1. Identify assets
 - 2. Identify threats
 - 3. Identify vulnerabilities
 - 4. Estimate impact

Risk Assessment

- Risk Assessment evaluates and prioritizes risks.
- Steps:
 - 1. Risk identification
 - 2. Risk evaluation
 - 3. Risk prioritization
 - 4. Risk mitigation

Risk Assessment Example

- Asset: Web Server
- Threat: DDoS Attack
- Vulnerability: Open ports
- Impact: Website downtime
- Risk Level: High

Network Port

- A port is a logical communication endpoint used by computers to identify services.
- Ports work with IP addresses to send and receive data.

Total Number of Ports

- There are 65,535 ports.
- Port Ranges:
 - - 0–1023: Well-known ports
 - - 1024–49151: Registered ports
 - - 49152–65535: Dynamic/Private ports

Working of Ports

- Ports help identify which service or application data belongs to.
- Example:
- IP Address: Identifies the device
- Port Number: Identifies the service

Common Ports and Services

- Port 20/21 – FTP (File Transfer)
- Port 22 – SSH (Secure Login)
- Port 23 – Telnet
- Port 25 – SMTP (Email)
- Port 53 – DNS
- Port 80 – HTTP
- Port 443 – HTTPS

HTTP & HTTPS

- HTTP (Port 80): Unsecured web communication
- HTTPS (Port 443): Secured web communication using SSL/TLS

HTTP Status Codes

- 200 – OK (Request successful)
- 301 – Moved Permanently
- 403 – Forbidden
- 404 – Not Found
- 500 – Internal Server Error

404 Error

- 404 Not Found means the server is reachable but the requested page does not exist.
- It is not a hacking error, but attackers can misuse misconfigured pages.

Role of Ports in Cyber Security

- - Identify running services
- - Help detect unauthorized access
- - Used in firewall rules
- - Important for intrusion detection

Open Ports as a Risk

- Open ports can be exploited by attackers.
- Example:
- Open Telnet (Port 23) can allow unauthorized access.

Port Scanning

- Port scanning checks which ports are open.
- Used by:
 - - Hackers (for attacks)
 - - Security teams (for defense)

Port Analysis Tools

- - Nmap
- - Netstat
- - Wireshark
- - Firewall logs

Practical Use of Ports

- - Close unused ports
- - Allow only required services
- - Monitor unusual port activity
- - Apply firewall rules

Summary

- Threats exploit vulnerabilities to create risks.
- Risk analysis and assessment help reduce damage.
- Ports play a critical role in networking and cyber security.